

SECTION 15941

AIR DEVICES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section includes grilles, diffusers, particulate control systems, and roof hoods associated with air distribution systems.

1.3 SUBMITTALS

- A. Submit in accordance with Division 1.
- B. Submit product data and shop drawings covering each item together with schedule of air devices.
- C. Submit manufacturer's installation instructions.
- D. Submit appropriate noise data.

1.4 QUALITY ASSURANCE

- A. Make airflow tests and sound level measurements in accordance with applicable ADC equipment test codes and ASHRAE standards.
- B. Manufacturer shall certify cataloged performance and ensure correct application of air device types.

1.5 COORDINATION

- A. Review requirements of air devices as to size, finish, and type of mounting prior to submitting shop drawings and air device schedules.
- B. Check location of outlets and make necessary adjustments in position to conform with architectural features, symmetry, and lighting arrangement.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. Air Devices:
 - 1. According to the recommended manufacturer table.
- B. Air Distribution and Airborne Particulate Control System:
 - 1. According to the recommended manufacturer table.
- C. Roof Hoods:
 - 1. According to the recommended manufacturer table.
- D. Perforated Duct Air Outlets: United McGill.
 - 1. According to the recommended manufacturer table.

2.2 GENERAL REQUIREMENTS

- A. Rate units in accordance with ADC standards.
- B. Base air device application on function-specific noise criteria.

Laboratories	NC-35
Computer Equipment	NC-35
Rooms	
Public Circulation	NC-35
Areas	
Open-plan Offices	NC-30
Library	NC-25
Executive and Private	NC-20
Offices	
Conference Rooms	NC-20
Auditorium	NC-10
- C. Provide supply devices with sponge rubber seal around edges.
- D. Provide baffles to direct air away from walls, columns, or other obstructions within radius of diffuser operation.
- E. Provide air device frames compatible with adjacent finish surfaces involved. Refer to Drawings and reflected ceiling plans.
 - 1. In Modular Ceilings: Snap-in T-bar, inverted T-bar or spline type as required.
 - 2. For Perforated Face Devices in Hard Ceilings: Curved, surface mounted.
 - 3. For All Other Devices in Hard Ceilings or Walls: Flat face frames, with 1-1/4 inches wide margins and countersunk screws.

4. Diffusers in Rough-textured Ceilings such as Acoustical Plaster: Provide antismudge frames.
5. Devices in Plaster Surfaces: Provide plaster frames.
6. Devices in Concrete Floors: Cast mounting frames in place.

F. Construction Change: Change aluminum air devices to all steel construction when utilized in conjunction with ceiling type fire dampers. Since ceiling type fire dampers are capable of balancing airflow, omit opposed blade dampers.

G. Performance: Refer to Drawings for air device types, sizes, and capacities.

2.3 SUPPLY GRILLES

A. Construct grilles using minimum 20 gauge extruded aluminum frames and individually adjustable streamlined vanes spaced on 3/4 inch centers. Keep vanes under tension to maintain set points.

2.4 INDUSTRIAL SUPPLY GRILLES

A. Construct grilles using minimum 20 gauge extruded aluminum frames and individually adjustable streamlined blades. Keep vanes under tension to maintain set points.

2.5 LINEAR SUPPLY GRILLES

A. Construct grilles using extruded aluminum components consisting of frame and fixed 1/8 inch by 3/4 inch grille bars.

2.6 LINEAR SUPPLY DIFFUSERS

- A. Diffusers: Construct of extruded aluminum, incorporating fully adjustable air pattern and flow control vanes capable of deflecting air from full horizontal to straight down or at any intermediate setting.
- B. Airflow Rate: Variable, without changing the air pattern. Provide vanes capable of controlling air patterns and flow rates without the use of an additional damper.
- C. Assembly: Install each length of diffuser without any visible means of fastening. Assemble multiple lengths with concealed aligning devices. Provide appropriate end caps.
- D. Diffuser Fastening Methods: Compatible with ceiling type used.

- E. Boots: TBSI insulated, provide where applicable for 1 or 2 slot diffusers in T-bar ceilings.

2.7 SQUARE OR RECTANGULAR DIFFUSERS

- A. Construct diffusers of welded steel square or rectangular, assembled in a louvered pattern with a 1, 2, 3, or 4 way throw, with each side delivering proportional air quantities to space. Provide removable cores.

2.8 PERFORATED FACE DIFFUSERS

- A. Construct diffusers of welded steel, with a removable perforated faceplate having concealed hinges and latches, removable diffuser core having curved adjustable blades capable of delivering air in 1, 2, 3, or 4-way pattern.

2.9 RETURN AND EXHAUST GRILLES

- A. Construct grilles with extruded aluminum frames and stationary vanes spaced on 3/4 inch centers.

2.10 GRID CORE RETURN AND EXHAUST GRILLES

- A. Construct grilles with extruded aluminum frames and grid core.

2.11 LINEAR RETURN OR EXHAUST GRILLES

- A. Construct grilles with extruded aluminum frames and fixed 1/8 inch by 3/4 inch grille bars.

2.12 PERFORATED RETURN OR EXHAUST GRILLES

- A. Construct grilles of steel, with removable perforated faceplate having concealed hinges and latches.

2.13 LIGHT TROFFER DIFFUSERS

- A. Light Troffer Diffusers: UL listed, constructed with 24 gauge zinc coated steel with slot areas painted flat black. Adjustable air pattern controllers and volume regulators, operable through fixture slots shall provide horizontal or vertical air patterns.
- B. Application: Use for both supply or return air, with flush or regressed flight fixture slots.
- C. Insulation: Provide supply air units with 1/2 inch fiberglass external insulated jacket covered with aluminum vapor barrier.
- D. Duct Connections: Dual top entry and single side entry, to match fixture types and lengths.

2.14 DOOR GRILLES

- A. Grilles: Construct using extruded aluminum double frames and V-shaped nonvision deflecting vanes spaced on 1/2 inch centers.
- B. Double Frames: 1-1/4 inches margins, adjustable to fit door thickness from 1-3/8 inches to 2 inches.

2.15 ROOF HOODS

- A. Air Inlet or Exhaust Hoods: Provide removable hood, curb flange and 1/2 inch mesh galvanized steel bird screen.
- B. Fabrication: Galvanized steel, minimum 16 gauge base and 20 gauge hood, or aluminum, minimum 16 gauge base and 18 gauge hood, suitably reinforced.
- C. Curb: Minimum 12 inches high with insulation between duct and curb.
- D. Finish: Factory baked enamel. Final finish shall be corrosion resistant epoxy paint, color selected by Architect.

2.16 GOOSENECKS

- A. Fabrication: Minimum 18 gauge galvanized steel, in accordance with SMACNA standards.
- B. Curb: Minimum 12 inch high curb base where size exceeds 9 inches by 9 inches.
- C. Finish: Prime coat and finish with corrosion resistant epoxy paint, color selected by Architect.

2.17 PERFORATED DUCT AIR OUTLETS

- A. Construct units of galvanized sheet steel, similar to manufactured round duct, circular cross-section, with perforations located within a portion of the circumference. Provide with or without a rotating damper as scheduled on Drawings.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install items in accordance with manufacturer's printed instructions.
- B. Paint ductwork visible behind air outlets matt black.

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